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Compilation:

Make

Running program:

Make test

Or

./schedule testcase1.txt FCFS

./schedule testcase1.txt SJF

./schedule testcase1.txt RR 5

Test cases:

Test case1:

4

0 0 12

1 2 4

2 3 1

3 4 2

```
akmekun@amadlaptop:~/CS4420/as5$ make test
./schedule testcase1.txt FCFS
```

```
=====
FCFS Scheduling Simulation
=====
```

```
Time 0 -> 1: PID 0
Time 1 -> 2: PID 0
Time 2 -> 3: PID 0
Time 3 -> 4: PID 0
Time 4 -> 5: PID 0
Time 5 -> 6: PID 0
Time 6 -> 7: PID 0
Time 7 -> 8: PID 0
Time 8 -> 9: PID 0
Time 9 -> 10: PID 0
Time 10 -> 11: PID 0
Time 11 -> 12: PID 0
PID 0 completed at time 12
Time 12 -> 13: PID 1
Time 13 -> 14: PID 1
Time 14 -> 15: PID 1
Time 15 -> 16: PID 1
PID 1 completed at time 16
Time 16 -> 17: PID 2
PID 2 completed at time 17
Time 17 -> 18: PID 3
Time 18 -> 19: PID 3
PID 3 completed at time 19
```

FCFS:

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	2	12	16	4	10
2	3	16	17	1	13
3	4	17	19	2	13

Average Waiting Time: 9.00

```
./schedule testcase1.txt SJF
```

```
=====
SJF Scheduling Simulation
=====
```

```
Time 0 -> 1: PID 0
Time 1 -> 2: PID 0
Time 2 -> 3: PID 0
Time 3 -> 4: PID 0
Time 4 -> 5: PID 0
Time 5 -> 6: PID 0
Time 6 -> 7: PID 0
Time 7 -> 8: PID 0
Time 8 -> 9: PID 0
Time 9 -> 10: PID 0
Time 10 -> 11: PID 0
Time 11 -> 12: PID 0
PID 0 completed at time 12
Time 12 -> 13: PID 2
PID 2 completed at time 13
Time 13 -> 14: PID 3
Time 14 -> 15: PID 3
PID 3 completed at time 15
Time 15 -> 16: PID 1
Time 16 -> 17: PID 1
Time 17 -> 18: PID 1
Time 18 -> 19: PID 1
PID 1 completed at time 19
```

SJF:

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
2	3	12	13	1	9
3	4	13	15	2	9
1	2	15	19	4	13

Average Waiting Time: 7.75

```
./schedule testcase1.txt RR 5
```

```
=====
```

```
Round Robin Scheduling Simulation
```

```
Time Quantum = 5
```

```
=====
```

```
Time 0: PID 0 entered ready queue
```

```
Time 0 -> 1: PID 0
```

```
Time 1 -> 2: PID 0
```

```
Time 2: PID 1 entered ready queue
```

```
Time 2 -> 3: PID 0
```

```
Time 3: PID 2 entered ready queue
```

```
Time 3 -> 4: PID 0
```

```
Time 4: PID 3 entered ready queue
```

```
Time 4 -> 5: PID 0
```

```
Time 5 -> 6: PID 1
```

```
Time 6 -> 7: PID 1
```

```
Time 7 -> 8: PID 1
```

```
Time 8 -> 9: PID 1
```

```
PID 1 completed at time 9
```

```
Time 9 -> 10: PID 2
```

```
PID 2 completed at time 10
```

```
Time 10 -> 11: PID 3
```

```
Time 11 -> 12: PID 3
```

```
PID 3 completed at time 12
```

```
Time 12 -> 13: PID 0
```

```
Time 13 -> 14: PID 0
```

```
Time 14 -> 15: PID 0
```

```
Time 15 -> 16: PID 0
```

```
Time 16 -> 17: PID 0
```

```
Time 17 -> 18: PID 0
```

```
Time 18 -> 19: PID 0
```

```
PID 0 completed at time 19
```

```
RR (Time quantum = 5):
```

PID	Start Time	End Time	Running Time
-----	------------	----------	--------------

-------	--	--	--

0	0	5	5
---	---	---	---

1	5	9	4
---	---	---	---

2	9	10	1
---	---	----	---

3	10	12	2
---	----	----	---

0	12	17	5
---	----	----	---

0	17	19	2
---	----	----	---

PID	Arrival Time	Running Time	End Time	Waiting Time
-----	--------------	--------------	----------	--------------

-------	--	--	--	--

0	0	12	19	7
---	---	----	----	---

1	2	4	9	3
---	---	---	---	---

2	3	1	10	6
---	---	---	----	---

3	4	2	12	6
---	---	---	----	---

```
Average Waiting Time: 5.50
```

Test case 2:

5

0 0 3

1 1 5

2 1 2

3 7 1

4 8 4

```
./schedule testcase2.txt FCFS
```

```
=====
FCFS Scheduling Simulation
=====
```

```
Time 0 -> 1: PID 0
Time 1 -> 2: PID 0
Time 2 -> 3: PID 0
PID 0 completed at time 3
Time 3 -> 4: PID 1
Time 4 -> 5: PID 1
Time 5 -> 6: PID 1
Time 6 -> 7: PID 1
Time 7 -> 8: PID 1
PID 1 completed at time 8
Time 8 -> 9: PID 2
Time 9 -> 10: PID 2
PID 2 completed at time 10
Time 10 -> 11: PID 3
PID 3 completed at time 11
Time 11 -> 12: PID 4
Time 12 -> 13: PID 4
Time 13 -> 14: PID 4
Time 14 -> 15: PID 4
PID 4 completed at time 15
```

```
FCFS:
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	3	3	0
1	1	3	8	5	2
2	1	8	10	2	7
3	7	10	11	1	3
4	8	11	15	4	3

```
Average Waiting Time: 3.00
```

```
./schedule testcase2.txt SJF
```

```
=====
SJF Scheduling Simulation
=====
```

```
Time 0 -> 1: PID 0
Time 1 -> 2: PID 0
Time 2 -> 3: PID 0
PID 0 completed at time 3
Time 3 -> 4: PID 2
Time 4 -> 5: PID 2
PID 2 completed at time 5
Time 5 -> 6: PID 1
Time 6 -> 7: PID 1
Time 7 -> 8: PID 1
Time 8 -> 9: PID 1
Time 9 -> 10: PID 1
PID 1 completed at time 10
Time 10 -> 11: PID 3
PID 3 completed at time 11
Time 11 -> 12: PID 4
Time 12 -> 13: PID 4
Time 13 -> 14: PID 4
Time 14 -> 15: PID 4
PID 4 completed at time 15
```

SJF:

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	3	3	0
2	1	3	5	2	2
1	1	5	10	5	4
3	7	10	11	1	3
4	8	11	15	4	3

Average Waiting Time: 2.40

```
./schedule testcase2.txt RR 2
```

```
=====
```

```
Round Robin Scheduling Simulation
```

```
Time Quantum = 2
```

```
=====
```

```
Time 0: PID 0 entered ready queue
```

```
Time 0 -> 1: PID 0
```

```
Time 1: PID 1 entered ready queue
```

```
Time 1: PID 2 entered ready queue
```

```
Time 1 -> 2: PID 0
```

```
Time 2 -> 3: PID 1
```

```
Time 3 -> 4: PID 1
```

```
Time 4 -> 5: PID 2
```

```
Time 5 -> 6: PID 2
```

```
PID 2 completed at time 6
```

```
Time 6 -> 7: PID 0
```

```
Time 7: PID 3 entered ready queue
```

```
PID 0 completed at time 7
```

```
Time 7 -> 8: PID 1
```

```
Time 8: PID 4 entered ready queue
```

```
Time 8 -> 9: PID 1
```

```
Time 9 -> 10: PID 3
```

```
PID 3 completed at time 10
```

```
Time 10 -> 11: PID 4
```

```
Time 11 -> 12: PID 4
```

```
Time 12 -> 13: PID 1
```

```
PID 1 completed at time 13
```

```
Time 13 -> 14: PID 4
```

```
Time 14 -> 15: PID 4
```

```
PID 4 completed at time 15
```

```
RR (Time quantum = 2):
```

PID	Start Time	End Time	Running Time
0	0	2	2
1	2	4	2
2	4	6	2
0	6	7	1
1	7	9	2
3	9	10	1
4	10	12	2
1	12	13	1
4	13	15	2

PID	Arrival Time	Running Time	End Time	Waiting Time
0	0	3	7	4
1	1	5	13	7
2	1	2	6	3
3	7	1	10	2
4	8	4	15	3

```
Average Waiting Time: 3.80
```

Test case 3:

6

0 0 6

1 0 2

2 0 8

3 0 3

4 0 1

5 0 4

```
./schedule testcase3.txt FCFS
```

```
=====
FCFS Scheduling Simulation
=====
```

```
Time 0 -> 1: PID 0
Time 1 -> 2: PID 0
Time 2 -> 3: PID 0
Time 3 -> 4: PID 0
Time 4 -> 5: PID 0
Time 5 -> 6: PID 0
PID 0 completed at time 6
Time 6 -> 7: PID 1
Time 7 -> 8: PID 1
PID 1 completed at time 8
Time 8 -> 9: PID 2
Time 9 -> 10: PID 2
Time 10 -> 11: PID 2
Time 11 -> 12: PID 2
Time 12 -> 13: PID 2
Time 13 -> 14: PID 2
Time 14 -> 15: PID 2
Time 15 -> 16: PID 2
PID 2 completed at time 16
Time 16 -> 17: PID 3
Time 17 -> 18: PID 3
Time 18 -> 19: PID 3
PID 3 completed at time 19
Time 19 -> 20: PID 4
PID 4 completed at time 20
Time 20 -> 21: PID 5
Time 21 -> 22: PID 5
Time 22 -> 23: PID 5
Time 23 -> 24: PID 5
PID 5 completed at time 24
```

FCFS:

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	6	6	0
1	0	6	8	2	6
2	0	8	16	8	8
3	0	16	19	3	16
4	0	19	20	1	19
5	0	20	24	4	20

Average Waiting Time: 11.50

```
./schedule testcase3.txt SJF
```

```
=====
SJF Scheduling Simulation
=====
```

```
Time 0 -> 1: PID 4
PID 4 completed at time 1
Time 1 -> 2: PID 1
Time 2 -> 3: PID 1
PID 1 completed at time 3
Time 3 -> 4: PID 3
Time 4 -> 5: PID 3
Time 5 -> 6: PID 3
PID 3 completed at time 6
Time 6 -> 7: PID 5
Time 7 -> 8: PID 5
Time 8 -> 9: PID 5
Time 9 -> 10: PID 5
PID 5 completed at time 10
Time 10 -> 11: PID 0
Time 11 -> 12: PID 0
Time 12 -> 13: PID 0
Time 13 -> 14: PID 0
Time 14 -> 15: PID 0
Time 15 -> 16: PID 0
PID 0 completed at time 16
Time 16 -> 17: PID 2
Time 17 -> 18: PID 2
Time 18 -> 19: PID 2
Time 19 -> 20: PID 2
Time 20 -> 21: PID 2
Time 21 -> 22: PID 2
Time 22 -> 23: PID 2
Time 23 -> 24: PID 2
PID 2 completed at time 24
```

```
SJF:
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
4	0	0	1	1	0
1	0	1	3	2	1
3	0	3	6	3	3
5	0	6	10	4	6
0	0	10	16	6	10
2	0	16	24	8	16

```
Average Waiting Time: 6.00
```

```
./schedule testcase3.txt RR 3
```

```
=====
```

```
Round Robin Scheduling Simulation
```

```
Time Quantum = 3
```

```
=====
```

```
Time 0: PID 0 entered ready queue
```

```
Time 0: PID 1 entered ready queue
```

```
Time 0: PID 2 entered ready queue
```

```
Time 0: PID 3 entered ready queue
```

```
Time 0: PID 4 entered ready queue
```

```
Time 0: PID 5 entered ready queue
```

```
Time 0 -> 1: PID 0
```

```
Time 1 -> 2: PID 0
```

```
Time 2 -> 3: PID 0
```

```
Time 3 -> 4: PID 1
```

```
Time 4 -> 5: PID 1
```

```
PID 1 completed at time 5
```

```
Time 5 -> 6: PID 2
```

```
Time 6 -> 7: PID 2
```

```
Time 7 -> 8: PID 2
```

```
Time 8 -> 9: PID 3
```

```
Time 9 -> 10: PID 3
```

```
Time 10 -> 11: PID 3
```

```
PID 3 completed at time 11
```

```
Time 11 -> 12: PID 4
```

```
PID 4 completed at time 12
```

```
Time 12 -> 13: PID 5
```

```
Time 13 -> 14: PID 5
```

```
Time 14 -> 15: PID 5
```

```
Time 15 -> 16: PID 0
```

```
Time 16 -> 17: PID 0
```

```
Time 17 -> 18: PID 0
```

```
PID 0 completed at time 18
```

```
Time 18 -> 19: PID 2
```

```
Time 19 -> 20: PID 2
```

```
Time 20 -> 21: PID 2
```

```
Time 21 -> 22: PID 5
```

```
PID 5 completed at time 22
```

```
Time 22 -> 23: PID 2
```

```
Time 23 -> 24: PID 2
```

```
PID 2 completed at time 24
```

```
RR (Time quantum = 3):
```

PID	Start Time	End Time	Running Time

0	0	3	3
1	3	5	2
2	5	8	3
3	8	11	3
4	11	12	1
5	12	15	3
0	15	18	3
2	18	21	3

```
5      21      22      1
2      22      24      2
```

PID	Arrival Time	Running Time	End Time	Waiting Time

0	0	6	18	12
1	0	2	5	3
2	0	8	24	16
3	0	3	11	8
4	0	1	12	11
5	0	4	22	18

Average Waiting Time: 11.33

❖ akmekun@amadlaptop:~/CS4420/as5\$